

Side-Effect Term Matching for Computational Adverse Drug Predictions

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Objectives

The establishment of polypharmacological networks, all the interactions between a collection of drugs and proteins, will allow for the exploration of drug re-purposing, side effect prediction, and the development of more efficacious drugs, targeting multiple proteins in a disease pathway. This project will help pave the way for the computational prediction of adverse drug reactions using a polypharmacological network built using molecular docking scores, an efficient prediction of how and how well drugs bind to a protein. The research presented here is an effort to map side-effect terms associated with a toxicity screen¹, known proteins in which drugs should not interact, to known side-effects associated with FDA-approved drugs² in order to test the accuracy of predictions.

Methods

1. Edit Distance: it is a methodology using which we can determine the minimum amount of string editing required transforming a source string into destination string. Editing operation includes insertion, deletion and substitution. Each operation involves a cost. The idea is to find out the minimum cost path from initial state to goal state. Where the initial state is the source string and goal state is the destination string. Now problem is using a naïve approach it is almost impossible to reach to a goal state from initial state as the space for edit sequences are quite large. Therefore we apply tabular approach to store the solution to sub problem to avoid recomputation that reduces the whole complexity from exponential to polynomial.

Following is the recurrence relation that we used to implement the dynamic version of edit distance algorithm.

- Initial state:
Memo(i,0) = i
Memo(0, j) = j
- For each in i= 1... length of source string S
For each in j = 1.....length of destination string D

$$\text{Memo}(i,j) = \begin{cases} D(i-1,j) + 1 \\ D(i,j-1) + 1 \\ D(i-1,j-1) + \begin{cases} 2 & \text{if } x(i) \neq y(j) \\ 0 & \text{if } x(i) = y(j) \end{cases} \end{cases}$$

- Final state
Memo(S,D) contains the transformation cost or total editing cost
2. Kuth Prat Morris: it is a pattern matching algorithm. It can find the number of times a given text pattern occurred within a text document. It does not generate any cost while trying to find out the pattern from documents. However we suitably modified the algorithm in order to get a cost while it is trying to find out a match of pattern within document text. We defined a cost function as follows:

$\text{matched_score} = \text{matched pattern length} / \text{document length} * 100$

Where matched pattern can be a complete pattern match or partial pattern match

3. Sense Disambiguation using meta map: This methodology takes the advantage of UMLS(Unified medical language system) Metathesaurus which is a large multi purpose thesaurus contains medical and health related concepts, concepts networks that is relation between different concepts and their synonyms . Meta map is an interactive tool that can generate concept cuis for medical terms. Now related medical terms will have same concept cuis. Using meta map tool we can get the list of concepts which are related to a side effect name. In this way for each side effect we will get a set of concept cuis. Idea is if we take the intersection of concepts set of any two side effects and it generate a non empty set that means those two side effects are related to each other through concept network or concept relationship.
4. Using Language Model: a language model works like this, given a partially formed sentence, the algorithm will try to find out the most appropriate word or words to form the full and proper sentence. We need a large dataset to train a language model in order to get a reasonable result while we are testing. The algorithm maps each word from train documents into a n dimensional Euclidean space. As a result each word becomes a n dimensional vector at the end of training process. We have used the pubmed journals, abstract, titles to train the language model which is actually a neural network modified to handle large dataset. We have considered a sentence context size of 10 while we trained the neural network and the generated vector size 300.

Results

Method 1

Vitro Profiled side effect name	FDR side effect name	Editing cost
psychosis schizophrenia	schizophrenia	11
muscle relaxation	muscle relaxation	0
sexual dysfunction	sexual dysfunction	0

Table I

Method2

Vitro Profiled side effect name	FDR side effect name	Editing cost
Hyperglycaemia	hyperglycaemia	93.3333333333
prostate carcinoma	prostate carcinoma	94.7368421053
cardiac hypertrophy	cardiac hypertrophy	95.0

Table II

Method 3

Vitro Profiled side effect name	FDR side effect name	Preferred Name	Concept cui
Potential cardiac valvulopathy	cardiac valve disease	heart valve disease	c0018824
cardiac hypertrophy	cardiac hypertrophy	cardiac hypertrophy	c1383860
in body weight	weight decreased	body weight	c0005910

Table III

Method 4

Vitro Profiled side effect name	FDR side effect name	Max score	Min score
cerebral and coronary artery vasoconstriction	Coronary artery disease	8.33754030863	-1.0
gastric emptying	Impaired gastric emptying	6.83925867081	-1.0
gallbladder contraction	Gallbladder disorder	9.01812928915	-1.0

Table IV

Methods are still being refined in order to achieve the maximum number of terms matched with a high-degree of accuracy.

Conclusion

This significant research will help alleviate the current economic burden of developing new pharmaceuticals by innovatively utilizing massive computational power. This research will lead to the establishment of structures to use in virtual drug screenings that will predict ADRs quickly and efficiently, resulting in safer clinical trials, as fewer drugs with negative off-target effects will advance to this stage, and more affordable therapies, as drugs destined for failure will be predicted earlier in the drug discovery process. This project will address important public health concerns by providing safer and more affordable drugs.

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